

The Cliffs of Dover

One of England's most famous natural landmarks, a parade of tall, brilliant white chalk cliffs, facing in the direction of France



▽ **WHITER THAN WHITE**
The cliffs owe their gleaming whiteness to the almost 100 percent purity of the chalk.

The Cliffs of Dover, which form part of the coast of the English county of Kent, run along the northwestern side of the narrowest part of the English Channel. They are made from chalk formed during the Cretaceous period, between 100 and 70 million years ago. At that time, a large part of what is now northwestern Europe was underwater. The shells of tiny marine organisms accumulated on the sea floor and became compressed into a layer of chalk hundreds of feet thick. Later, after sea levels fell, this chalk formed a land bridge between Britain and mainland Europe. Around 8,500 years ago, this bridge was swept away in a catastrophic flood, leaving behind the cliffs of Dover and similar white cliffs at Cap Blanc Nez in France.



▷ **BRILLIANT BLUE**
The Adonis Blue butterfly favors chalk downland, such as the area of Kent behind the cliffs, and is found there in modest numbers.



Fossils found in the **cliffs** include **sharks' teeth, sponges, and corals**

The Dune du Pilat

The tallest sand dune in Europe, located close to Arcachon Bay in southwestern France, with magnificent views from the top



EVOLUTION OF THE DUNE

Analysis of sand samples from the dune and the underlying soil has shown how the dune has grown over several centuries. Wind power is currently moving it a few feet per year toward the forest to its east.

5th-17th century: 65-130 ft (20-40 m) high	18th-19th century: 230-260 ft (70-80 m) high
2,000-500 BCE: 10-12 ft (3-4 m) high	today: 360 ft (110 m) high
8th-3rd century BCE: only slightly above sea level	

Containing more than 2 billion cubic feet (60 million cubic meters) of sand and running parallel to the shoreline, the Dune du Pilat rises gently to a height of 360 ft (110 m) above the west coast of France. It is nearly 2 miles (3km) long and 1,600 ft (500 m) wide. The word

Pilat comes from the Gascon word *Pilhar*, which means “heap” or “mound.” It takes about 30 minutes for a fit person to climb to the dune’s top, where the wind can be very strong. It is a great vantage point in fall for viewing flocks of migratory birds.



◁ **SHIFTING SAND**
The dune is slowly migrating away from the sea. Unless this movement can be stopped, the sand will eventually engulf a small forest on its land side.